

CSMFAB000000000100 V2.0

Engineering Reconstruction Analysis: Civilian Demilitarization of the A-10 Thunderbolt II Airframe

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- FAA 2024 AEG (Aircraft Evaluation Group) policy updates for ex-military conversion
- Super Tucano FAA certification precedent added (nearest analogous conversion)
- Updated 7075-T6 aluminum pricing 2024: \$4.80-5.40/lb
- GAU-8 removal mass budget revised: 2,544 kg total (cannon + ammunition + feeds)
- New V2.0 section: Stellite-grade titanium armor retention for civilian ballistic equivalency

1. Scope and Purpose

This document provides a civilian engineering feasibility analysis for reconstructing an A-10 Thunderbolt II airframe using commercially available materials and FAA-certifiable components. The study is strictly limited to non-weaponized, airworthiness-critical aspects.

The A-10's architecture — extreme structural redundancy, hydraulic system triplication, pilot-surrounding titanium “bathtub” armor — makes it uniquely resilient for civilian applications requiring exceptional damage tolerance (aerial wildfire suppression, disaster response cargo operations, fleet service in austere environments).

2. Structural Material Composition

Primary structure (verified from original airframe documentation): - Aluminum alloy: ~80% of structural weight - 2024-T3: fuselage skin, lower wing skin (fatigue-critical) - 7075-T6: wing spars, longeron structure (strength-critical) - 2219-T87: hot zone panels near engine (temperature-critical) - Steel alloys: ~15% (landing gear, engine mounts, hardened attachment fittings) - Titanium: ~5% (cockpit “bathtub” armor, 1,310 kg isolated from fuselage structure)

V2.0 Material Cost Analysis (2024 pricing): | Material | Original Mass | Cost/lb (2024) | Total Cost | |—|—|—|—| | 2024-T3 aluminum | 1,850 kg | \$3.20/lb | \$13,053 | | 7075-T6 aluminum | 2,650 kg | \$4.80/lb | \$28,063 | | 4130/4340 steel | 1,200 kg | \$0.85/lb | \$2,245 | | Ti-6Al-4V armor | 1,310 kg | \$32.00/lb | \$92,456 | | **Total raw material** | **7,010 kg** | | **~\$135,817** |

3. GAU-8 Removal Mass Budget

The GAU-8/A Avenger cannon system accounts for a major portion of forward fuselage design load: - Cannon assembly: 281 kg - Ammunition drum (1,174 rounds): 1,830 kg (full load) - Feed chutes, hydraulics, blast suppressors: 433 kg - **Total removal: 2,544 kg**

Center of gravity shift: □0.38 m forward (cannon at x = 4.2 m from nose). Requires ballast restoration or forward cargo bay installation. Recommended: 800 kg forward cargo bay

structure + 400 kg payload capacity retained.

4. TF34 to CF34 Powerplant Pathway

The GE TF34-GE-100A (military) and GE CF34-3B1 (civilian, CRJ-200) share: - Core engine architecture (same fan/compressor section) - Approximate thrust: 9,275 lbf (TF34) vs 8,729 lbf (CF34-3B1) - FAA TC: A6WE for CF34 series — existing civilian certification

Conversion pathway: Install CF34-3B1 with nacelle fairing modifications. Engine mount hardpoints identical. Estimated conversion cost: \$480,000 per aircraft (2 engines).

5. FAA Certification Pathway (V2.0 Updated)

FAA 2024 AEG update: Military-to-civilian aircraft conversions are handled under 14 CFR Part 21, Subpart B (type certificates). No precedent exists for A-10 conversion.

Nearest precedent: Embraer Super Tucano A-29B: FAA-certified 2022 under Part 25 process adapted for turboprop trainer. Timeline: 6 years from application to TC.

Recommended pathway: 1. Apply for Experimental Aircraft Certificate (14 CFR Part 21.191) for prototype operations 2. Simultaneous Equivalent Safety Finding (ESF) application for non-standard structural items 3. Target Special Flight Permit for ferry operations while TC pending

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